CSA5108 Cryptography and Network security

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Exp 4: Polyalphabetic Substitution Cipher - Vigenere Cipher
Code:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
   char text[1000], key[100];
   int i, j = 0, shift;

   printf("Enter plaintext: ");
   fgets(text, sizeof(text), stdin);

   printf("Enter key: ");
   scanf("%s", key);

   for (i = 0; text[i] != '\0'; i++) {
      if (isalpha(text[i])) {
         shift = toupper(key[j % strlen(key)]) - 'A';

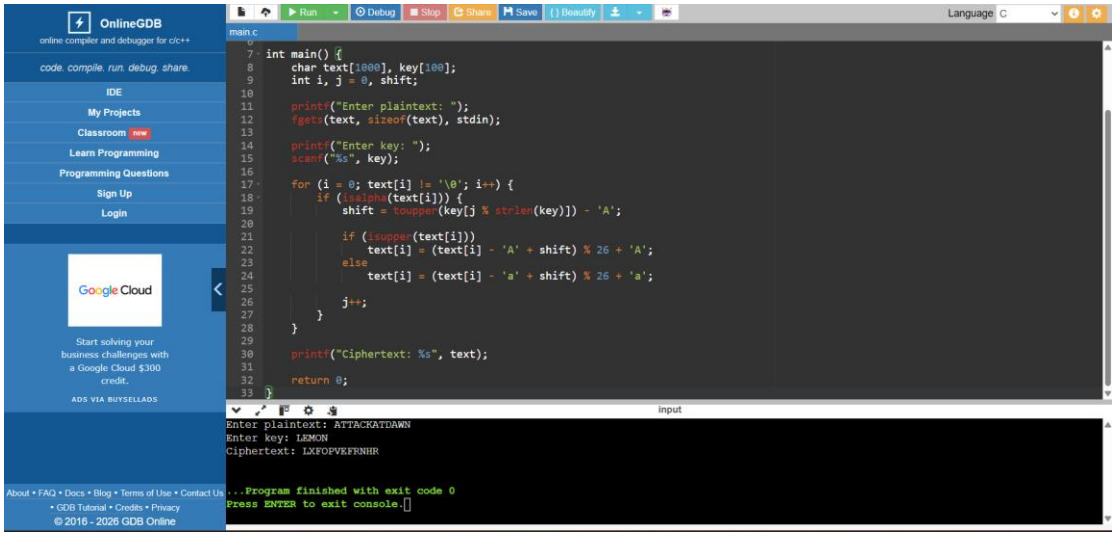
         if (isupper(text[i]))
            text[i] = (text[i] - 'A' + shift) % 26 + 'A';
         else
            text[i] = (text[i] - 'a' + shift) % 26 + 'a';

         j++;
      }
   }

   printf("Ciphertext: %s", text);

   return 0;
```

}


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Language C
main.c
int main() {
char text[1000], key[100];
int i, j = 0, shift;
printf("Enter plaintext: ");
fgets(text, sizeof(text), stdin);
printf("Enter key: ");
scanf("%s", key);
for (i = 0; text[i] != '\0'; i++) {
if (isalpha(text[i])) {
shift = toupper(key[j % strlen(key)]) - 'A';
if (isupper(text[i]))
text[i] = (text[i] - 'A' + shift) % 26 + 'A';
else
text[i] = (text[i] - 'a' + shift) % 26 + 'a';
j++;
}
}
printf("Ciphertext: %s", text);
return 0;
}
input
Enter plaintext: ATTACKATDAWN
Enter key: LEMON
Ciphertext: LXFOPVEFRNHR
...Program finished with exit code 0
Press ENTER to exit console.